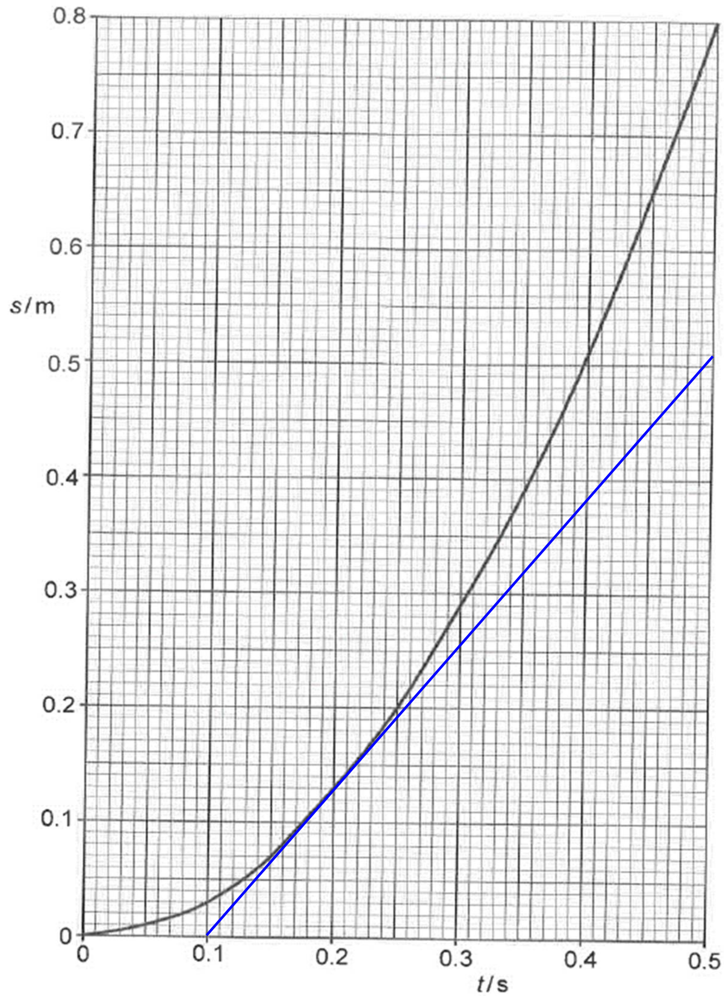
No.	Solution
1(a)	The block slides down the slope with uniform acceleration resulting in its velocity increasing at a constant rate.
1(b)(i)	 **Fig. 1.2** Take gradient coordinates of tangent (0.100, 0) and (0.500, 0.510) on Fig. 1.2. Speed *v* is the gradient of the tangent at *t* = 0.20 s. $v = \dfrac{0.510 - 0}{0.500 - 0.100}$ $= \dfrac{0.510}{0.400}$ $= 1.275$ $\approx 1.28 \text{ m s}^{-1}$

No.	Solution
1(b)(ii)	Using $v = u + at$, $a = \dfrac{v-u}{t}$ $\quad = \dfrac{1.275-0}{0.20}$ $\quad = 6.375$ $\quad \approx 6.38 \text{ m s}^{-2}$
2(a)	From Fig. 2.1, momentum of ball as it makes contact with the surface